

Methodology Note: Multi-Agent Reinforcement Learning Training

Starfarm project, decision-making component (farmers) *Document intended for review by RL experts. July 2026*

This note describes the **training methodology only**. The underlying simulator (a GAMA agronomic model) is treated as a black box: only the learning interface and the algorithmic choices are exposed here. The goal is to allow a judgement on the **coherence and quality** of the training loop.

1. Problem formalisation

The problem is cast as a **partially observable multi-agent Markov game (Dec-POMDP)**:

- **N homogeneous agents** (farmers) acting in parallel.
- **Partial observability**: each agent observes its local state plus aggregate market signals, never the full global state.
- **Market coupling across agents**: the price is endogenous (overproducing a premium good drives its price down). Rewards are therefore **neither additive nor stationary** from the point of view of an isolated agent, which motivates a centralized critic.
- **Finite horizon**: 25 decision steps (years), termination by truncation; $\gamma = 0.97$.
- **Configurable reward**: cooperative (global profit) or individual (own profit), plus two alternative objectives (minimising an environmental cost; a weighted combination). The mode affects only the learning signal, not the evaluation metric.

2. Temporal abstraction (semi-MDP)

- **One RL decision equals one simulated year**: farming practices are chosen annually. Actions are resampled only at year boundaries.
- **Dense reward per seasonal sub-step (about 122 days)**: a semi-MDP formulation. Decisions remain annual, but the critic receives a **dense intra-year** value signal, which reduces the variance of the value estimate.
- **GAE with a variable discount factor**: a sub-step covering d days discounts by $\gamma^{(d/365)}$, so that discounting stays consistent despite the variable granularity.
- *(Simulator engineering, outside RL semantics: steps are grouped into a batched macro-step to limit round-trips with the simulator; no effect on learning.)*

3. Spaces

Observation: Box(18), per agent, normalised online

Block	Dim.	Content
Local state	4	previous annual profit, mean fertiliser, soil health, yield

Local environment	2	pollution, salinity
Static descriptors	2	number of plots, area; enable specialisation of a shared actor
Market / coordination signals	3	premium share, relative premium price, fraction of agents already decided
Last action	6	memory of the previous decision
Time	1	normalised year index (remaining horizon)

The deliberate absence of **any agent identifier** makes the actor invariant to the number of agents.

Action: Box(6), continuous in $[0, 1]$

A continuous vector decoded into practices (one binary selector per practice via a 0.5 threshold, plus one continuous quantity for irrigation). The policy is a **diagonal Gaussian**; the $[0, 1]$ bound is enforced simulator-side by clipping (see section 12, known limitation).

Reward: 4 modes

global profit, per-agent profit, -pollution, profit - lambda * pollution. The **evaluation metric** (true global profit and pollution) is computed and exposed separately from the learning signal, which makes the modes directly comparable.

4. Sequential intra-year decision protocol

Each year, agents decide **one at a time, in a random order**. Each agent observes the market state **already updated** by the commitments of the agents that decided before it (dynamic premium share, decided fraction). Rationale: a shared deterministic policy is symmetric; on a market with decreasing returns (a minority-game / El Farol dynamic), that symmetry prevents diversification. The sequential order provides the **symmetry-breaking** mechanism without resorting to an agent identifier.

5. Algorithm: MAPPO (CTDE)

A **centralized training, decentralized execution** framework:

- **Decentralized shared-weight actor**: acts on the local observation only; execution is independent of N.
- **Centralized critic**: receives the **global state** (concatenation of all agents' normalised observations); used at training time only.
- **Clipped, on-policy PPO** optimisation (clip 0.2).

6. Architectures

Network	Body	Head	Notes
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Actor (shared)	MLP 2 x 128, tanh	linear mean + learned <code>log_std</code>	global <code>log_std</code> (init -1.2), state-independent; diagonal Normal
Critic (centralized)	MLP 2 x 256, tanh	scalar	input = <code>obs_dim</code> x N

A single **Adam** optimiser over the actor and critic parameters.

7. Advantage estimation and value targets

- **GAE(`lambda` = 0.95)** with per-sub-step variable discounting.
- **Per-decision advantage:** GAE evaluated at the **first sub-step** of the year, intended to summarise the discounted future from the decision point.
- **Two credit regimes:** (i) team reward, one shared advantage per time step; (ii) per-agent reward, the centralized critic predicts the **mean per-agent return**, and the individual advantage is the own return relative to that shared baseline.
- **Dual-batch update:** the **actor** learns only on **decision** transitions (annual); the **critic** learns on **all sub-steps** (dense target), at identical simulator cost.
- **Advantage normalisation** per mini-batch.

8. Stabilisation and regularisation

- **Observation normalisation:** running mean and variance (Welford).
- **Adaptive value-target normalisation:** a running scalar standard deviation, replacing a fixed reward-scale factor. The critic learns on normalised targets; the value is de-normalised before the GAE computation (advantages stay in raw scale).
- **Entropy decay:** coefficient annealed linearly (0.01 to 0.001).
- **KL early stopping:** actor epochs stop when `approx_kl > 1.5 * target_kl`; the critic completes its full cycle.
- **Gradient clipping** (norm 0.5), separate for actor and critic.
- **Linear learning-rate annealing** over the training run.

9. Hyperparameters

Parameter	Value	Parameter	Value
Algorithm	MAPPO (CTDE)	<code>gamma</code>	0.97
<code>gae_lambda</code>	0.95	PPO clip	0.2
epochs / update (<code>k_epochs</code>)	4	mini-batch	64
<code>lr</code>	3e-4 (annealed to 0)	<code>target_kl</code>	0.03
entropy coef	0.01 to 0.001	<code>value_coef</code>	0.5
actor width	128	critic width	256
<code>obs_dim</code>	18	<code>act_dim</code>	6

N (training)	10	sub-step	122 days
horizon	25 years	sequential decisions	yes
obs normalisation	yes	seed	0
episodes	400 to 600 per run	grad clip	0.5

10. Evaluation, checkpointing, generalisation

- **Periodic greedy evaluation** (every 25 episodes): a deterministic policy (the Gaussian mean) measuring the true objective. Used as the best-checkpoint selection criterion.
- **True objective always exposed** (profit and pollution) regardless of the training mode, giving a fair cross-mode comparison.
- **Determinism**: the environment reset is deterministic (reproducible year lengths); the inter-episode variance of the global profit is low (CV about 0.4 percent), most of the randomness coming from the sequential decision order.
- **Scale generalisation**: because the actor carries no agent identifier, a policy trained on 10 agents was transferred as is to **748 agents without retraining** (decentralized execution), which served as a robustness test.
- **Complete checkpoints**: actor and critic weights, optimiser state, observation and value normalisation statistics, allowing exact resumption.

11. Structural decisions and rationale

Decision	Rationale
Shared actor with no agent identifier	invariance to the number of agents and scale transfer; symmetry-breaking is provided by the sequential protocol, not by an identifier
CTDE / centralized critic	market coupling makes the environment non-stationary for an independent critic; the global critic improves credit assignment
Continuous <code>Box</code> actions (vs <code>MultiDiscrete</code>)	smoother space, a single Gaussian head, tunable irrigation quantity
Semi-MDP with dense reward	reduces critic variance without changing the annual decision granularity
Adaptive value normalisation	return magnitudes vary strongly across reward modes; a fixed scale factor would be miscalibrated

12. Known limitations and points submitted for review

1. **Unbounded Gaussian policy**, bounded by simulator-side clipping: a gradient bias at the bounds. A `tanh` squashing with log-probability correction would be more rigorous.
2. **Best-checkpoint selection by greedy profit**, including for non-profit objectives: the "best" checkpoint may be misaligned with the objective actually trained. Worked around by evaluating the final checkpoints for those objectives.

3. **Single greedy episode per policy** at evaluation: profit variance is low and quantified (CV about 0.4 percent), but the variance of secondary objectives is not quantified.
4. **State-independent `log_std`**: homogeneous, non-adaptive exploration.
5. **Degenerate local optimum** observed for the single-objective "pollution only" reward (exploration collapse once the profit signal is removed), suggesting that exploration depends on the shaping provided by profit.
6. **On-policy (PPO)**: limited sample efficiency, since each episode requires a full, costly rollout of the simulator.
7. **Reproducibility**: a single seed, no multi-seed confidence intervals; the configuration files are not versioned.
8. **Intra-year credit assignment**: the decision advantage is taken at the first sub-step; the assumption that this point summarises the discounted future has not been validated by ablation.
9. **KL early stopping** occasionally triggered on the very first mini-batch (observation normalisation drift between collection and update): cosmetic, self-correcting.

Available on request: training logs (per-episode returns, periodic greedy evaluations), convergence curves, and the source code of the training loop.